

# **STAREHE GIRLS' 2025 PREDICTION**



## **POSSIBLE KCSE EXAM**

232/3

## **PHYSICS**

### **PAPER 3 (PRACTICAL)**

**TIME: 2½ HOURS**



**NAME.....**

**SCHOOL..... SIGN.....**

**INDEX NO..... ADM NO.....**

*(Kenya Certificate of Secondary Education)*

### **INSTRUCTIONS TO CANDIDATES:-**

- Write your **name** and **Adm number** and **class** in the spaces provided above.
- Answer **all** the questions in the spaces provided in the question paper
- You are supposed to spend the first **15 minutes** of 2 ½ hours reading the whole paper carefully before commencing your work.
- Marks are given for a clear record of the observations actually made, their suitability, accuracy and the use of them
- Record your observations as soon as you make them.
- Mathematical tables, slide rules and silent non-programmable electronic calculators may be used.
- Take  $g = 10\text{ms}^{-2}$

### **For Examiners' Use Only:**

| QUESTION    | MAXIMUM<br>SCORE | CANDIDATE<br>SCORE |
|-------------|------------------|--------------------|
| 1           | 20               |                    |
| 2           | 20               |                    |
| TOTAL SCORE | 40               |                    |

## **QUESTION 1**

1 You are provided with the following.

- A new size D dry cell 1.5V in a cell holder
- A voltmeter (Range 0 – 2.5V or 0 – 3.0V)
- An ammeter (Range 0 – 1.0A)
- A Constantine wire, W, mounted on a millimeter scale on a wooden plank
- 7 connecting wires with at least one with a crocodile clip at one end
- A micrometer screw gauge (to be shared)
- A switch

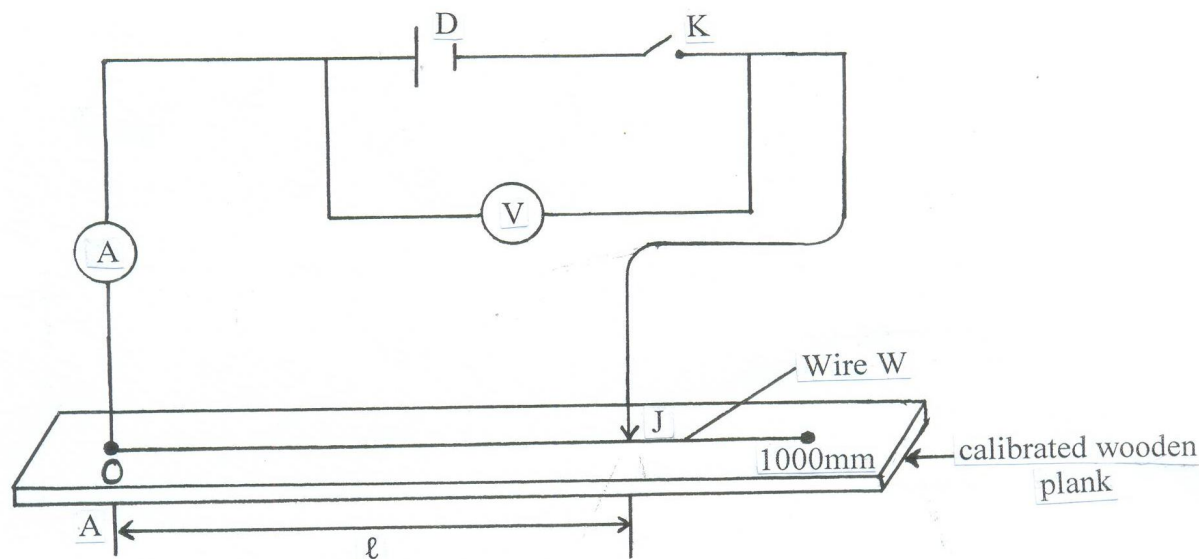
### **Proceed as follows:**

(a)( i) Measure the thickness,  $t$ , in meters of the wire, W,

$t =$  \_\_\_\_\_ (1mark)

(ii) Connect the circuit as shown in the diagram below.

NB: Ensure the circuit is complete before commencing the experiment. The switch K should control both circuits.



(b) Starting with the crocodile clip, J, at  $L = 200\text{mm}$  from A, close the switch K and record the voltmeter reading V and the corresponding ammeter reading I.

(i) Voltmeter reading,  $V =$  \_\_\_\_\_ (1mark)

(ii) Ammeter reading,  $I =$  \_\_\_\_\_ (1mark)

## **NOTE**

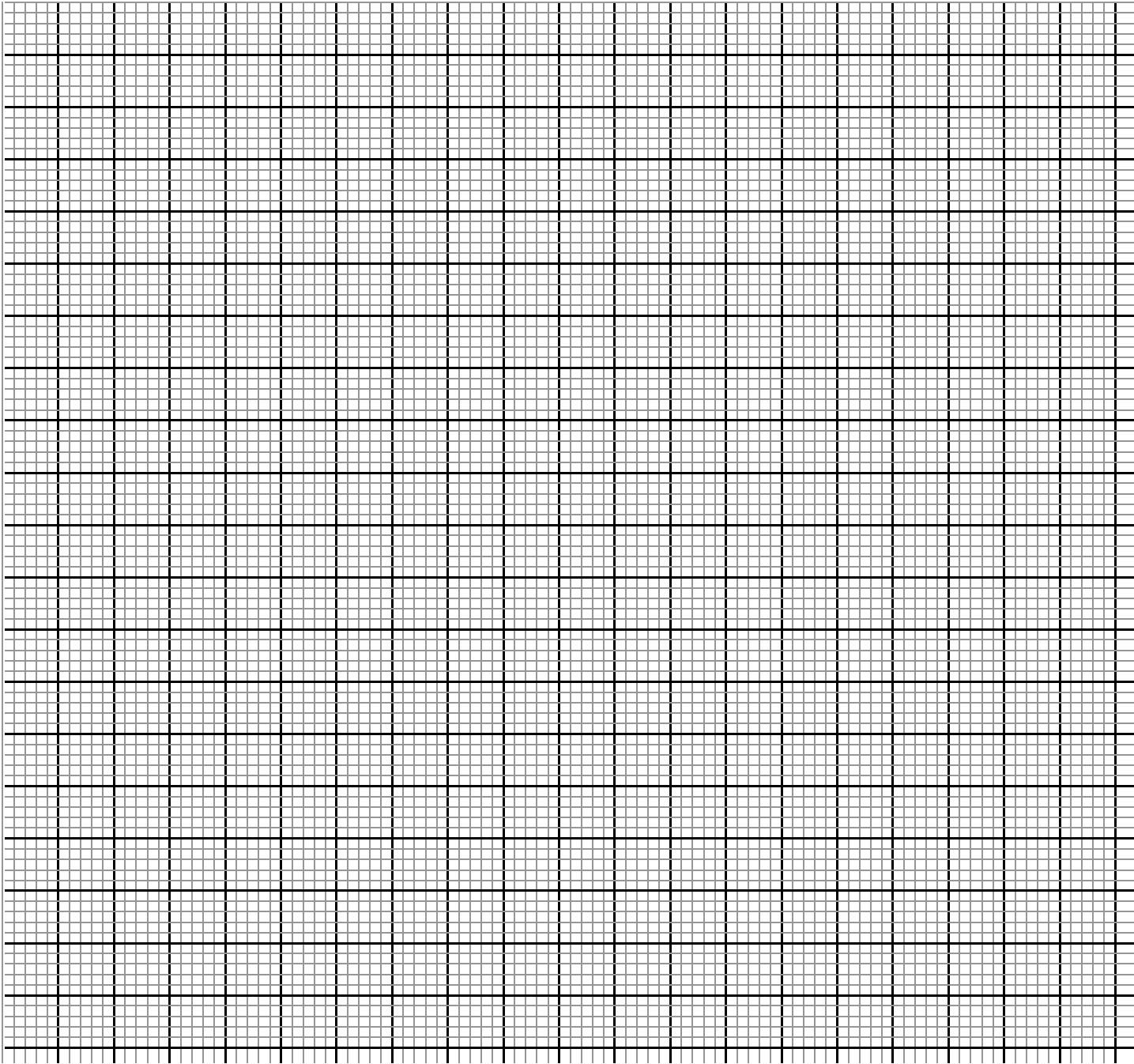
Open the switch, K, when not taking the readings

(c) (i) Repeat the procedure in (b) above for values of (L= 300, 400, 500, 600 and 800mm.

(ii) Record your results in the table below (5 marks)

|                        |     |     |     |     |     |     |
|------------------------|-----|-----|-----|-----|-----|-----|
| Length (AJ) L(mm)      | 200 | 300 | 400 | 500 | 600 | 700 |
| Voltmeter reading V(V) |     |     |     |     |     |     |
| Ammeter reading, I(A)  |     |     |     |     |     |     |
| $V/I$                  |     |     |     |     |     |     |

(d) Plot a graph of the voltmeter reading, V/I, (vertical axis) against Length L (m). (5marks)



(e) From your graph;

(i) Determine the slope, **S**,

**(3marks)**

(g) Now connect the voltmeter across the wire, **W**, to enable you obtain a potential drop across any part of length, **AJ**, of the wire, **W**

(i) Using the length, **AJ**, = **L** = 550mm, close the switch and then read the voltmeter and corresponding ammeter readings.

Voltmeter reading, **V** = \_\_\_\_\_ **V (1 mark)**

**Ammeter reading, I** = \_\_\_\_\_ **A (1 Mark)**

(j) Calculate the value of **P** from

$$P = 0.785 \frac{V t^2}{I L}$$

**I L**

Where **L**, **V**, **t** and **I** are quantities obtained in their SI units.

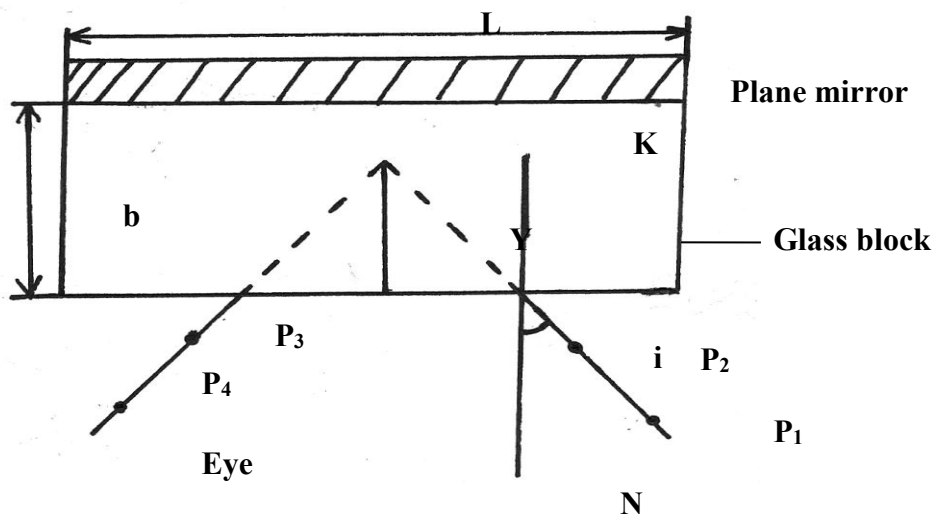
**(2marks)**

## **QUESTION 2 PART A**

You are provided with the following

- Rectangular glass block
- Five optical pins
- Half meter Rule
- Soft board
- Two sheets of Plain papers
- Plane mirror
- Cello tape
- Vernier calipers (to be shared)
- Four Thumb tacks or office pins

(e) Set up the apparatus as shown in figure below



Proceed as follows

a)(i) Using the vernier calipers provided, measure and record the breadth **b** of the glass block

**b**= \_\_\_\_\_ cm (1 mark)

(ii) Trace the outline of the glass block on the plain sheet of paper (**mounted on the soft board using office pins or thumb tacks**). Using cello tape, fix the mirror on one side along the length of the glass block.

(iii) Draw the normal **NK** on the plane paper along the outline as shown in the diagram and measure angle  **$i=20^\circ$**  from the normal

(iv) Draw the line representing the incident ray and fix pins **P1** and **P2** as shown in the figure

(v) By observing the images of the pins **P1** and **P2**, fix **P3** and **P4** such that they appear to be in a straight line with the images of **P1** and **P2** as seen in the glass block.

(vi) Join the points **P3** and **P4** and extend them to intersect line **P1 P2** produced. Measure the perpendicular distance **Y**

(vii) Repeat steps **(iii to vi)** for different values of **i** given and record your values in the table below

**2mks**

|           |    |    |    |    |
|-----------|----|----|----|----|
| $i^\circ$ | 20 | 25 | 30 | 35 |
| Y(cm)     |    |    |    |    |

**(viii)** Determine the average of the values of **Y**

**(2 marks)**

(ix) Determine the values of constant  $k$  given that  $k = \frac{b}{Y}$  (2marks)

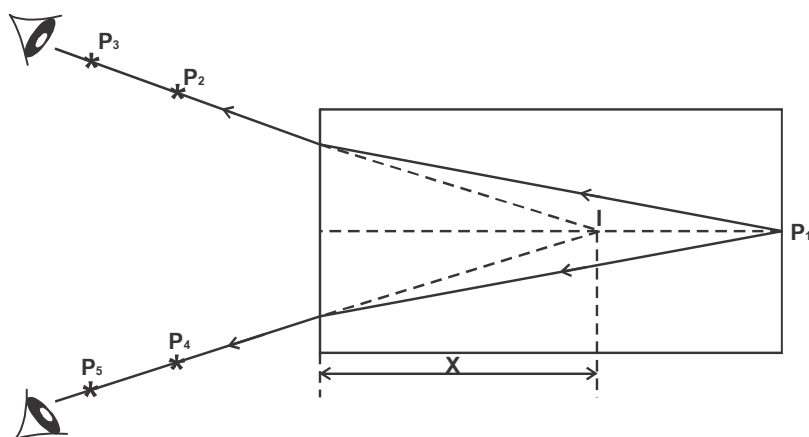
b) Measure the length,  $l$ , of the glass block using a half meter rule

$l = \dots\dots\dots$  cm

(1 mark)

c) Place the glass block on the second plain sheet of paper (**mounted on the soft board using office pins or thumb tacks**) and trace its outline.

d) Fix a pin  $P_1$  firmly at the middle of one end of the glass block as shown in figure below.



e) With your eye at the opposite end of the block, place pins  $P_2$  and  $P_3$  so that they are in line with the image  $I$  of pin  $P_1$  as shown in the figure above. Similarly locate the image  $I$  using pins  $P_4$  and  $P_5$ .

f) Remove the glass block and produce  $P_3P_2$  and  $P_5P_4$  to their point of intersection.

g) (i) Measure the distance,  $x$ , as shown in figure above using a half meter rule

$x = \dots\dots\dots$ cm (1 mark)

(ii) Determine the constant  $q$  given that

$q = l/x$  (2 marks)

iii) how does **q** compare with **k** in question a part ( ix) above  
(1/2 mark)

(iv) What does constant **k** and **q** represent?  
(1/2 mark)

### **QUESTION 2 PART B**

You are provided with the following:

- a boiling tube
- a measuring cylinders
- a half metre rule
- water in a container
- a stand complete with boss and clamp
- vernier callipers (may be shared)

**Proceed as follows:**

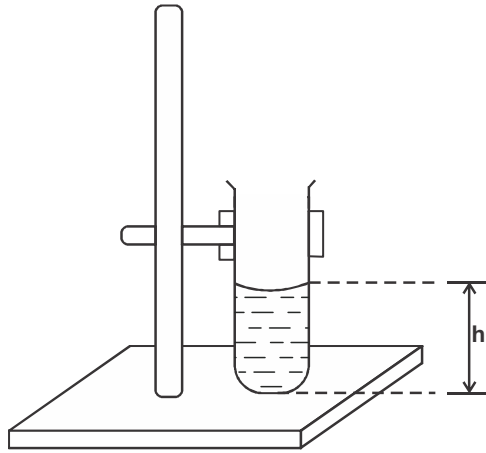
a) Using the vernier callipers measure the internal diameter, **d**, of the boiling tube.

(i) **d** = ..... cm. (1 mark)

(ii) Calculate the cross-sectional **A** area of the boiling **tube**

(2mks)

b) i) Clamp the boiling tube vertically as shown in the figure below.



**ii)** Using the measuring cylinder pour  $15\text{cm}^3$  of water into the boiling tube. Measure and record in the table 1 the height  $h$ , of water in the boiling tube.

**iii)** Repeat the procedure in b(ii) for other volumes of water,  $V$ , shown in the table.

| Volume, $V$ , of water<br>( $\text{cm}^3$ ) | Height, $h$ , of water<br>(cm) | $Ah$ |
|---------------------------------------------|--------------------------------|------|
| 15                                          |                                |      |
| 20                                          |                                |      |
| 25                                          |                                |      |
| 30                                          |                                |      |

**(3 marks)**

**iv)** Given that the volume of the liquid is  $V$  and height of the liquid is  $h$ , write an expression relating the volume, the height, and the area  $A$  of the boiling tube as indicated by the values in the table **above**

**(2 marks)**